

Moving Response Fibres: A Geometric View of Behaviour and Learning

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Abstract

Neural networks can agree on a prospectively declared behaviour while differing in their internal realizations and capacity for subsequent learning. Here, behaviour denotes a finite response map fixed before evaluation, not the model’s complete input-output function. We model the corresponding freedom by the state-dependent response kernel $\mathcal{V}_\theta = \ker DR(\theta)$. Prospective CNN and ResNet audits show that selected kernel directions carry adaptive value, that current-kernel transport and source-fixed replay have distinct finite realizability-cost scaling, and that small fibre drift can be amplified by more than three orders of magnitude in active response residual. We then test a functional consequence at recorded AdamW nodes. Under prospectively frozen low-response budgets, candidate scales are selected on calibration data and evaluated on disjoint held-out data. The instantaneous response-kernel arm outperforms recorded-optimizer, source-kernel, half-path time-shifted-kernel and signed-permuted controls in 7/8 reduced ResNet seeds and 8/8 GPT-2 native-LoRA-B seeds; all 24 noninitial GPT-2 nodes pass the frozen ordering. Thus a moving response fibre can provide local stability–plasticity freedom rather than merely a geometric null space. The claim remains confined to declared finite parameter charts and discrete response budgets: ordinary optimizers are not shown to navigate the fibre naturally, and restricted action minimization remains distinct from realization-cost minimization.

1 Introduction

Neural networks can realize the same prospectively declared behaviour through many different parameter states. Yet agreement on that behaviour need not imply agreement in what the models can learn next. The resulting question is geometric: how can models with the same declared response possess different capacities for subsequent learning, and how does that freedom change along a training path?

Two established lines bear on this question. Optimization-theoretic accounts explain how overparameterization [18], landscape geometry [10, 14, 8, 5] and implicit bias [16] shape solution selection. Representational and information-geometric accounts explain linearized dynamics [13], parameter-space metrics [2] and feature formation [3]. Together they explain much about why training succeeds and what representations emerge.

They leave a specific gap. They do not by themselves distinguish inert redundancy from directions that preserve a declared response while carrying future adaptive value. Nor do endpoint- or loss-based accounts ordinarily separate a path functional used to order response-preserving realizations from the correction cost required to make each finite proposal actually satisfy the declared response. For the low-dimensional responses considered here, overparameterization often makes $\dim \ker DR(\theta) = N - \text{rank } DR(\theta)$ large, but kernel membership alone implies neither adaptive value nor selected motion. This paper studies two consequences of one state-dependent mechanism. First, it asks whether movement of the response kernel changes the finite cost of realizing response-preserving paths and whether small chart drift can be strongly amplified in response space. Second, at recorded optimizer nodes, it asks whether the instantaneous kernel provides greater held-out learning utility than historical, time-misaligned, unstructured and unprojected directions under the same prospectively frozen response budgets. Existing tools contain many of the individual ingredients — null spaces, transported subspaces, pullback metrics and natural gradients — but the present work combines them into a prospective audit of adaptive value, online fibre transport, finite realization cost and restricted path action. The required comparison is therefore pathwise rather than endpoint-only, and keeps geometric admissibility distinct from finite realization cost.

We therefore take the response fibre as the primary object. For a prospectively frozen external response R and a target output y , the realization fibre and its tangent space are

$$\mathcal{F}_y = R^{-1}(y), \quad T_{\theta}\mathcal{F}_y \simeq \ker DR(\theta). \quad (1)$$

Functional freedom is the capacity to change an internal realization while preserving a prospectively declared external response. Its directions are invisible to the response to first order, which is precisely why they are usually dismissed as redundancy; the empirical content of this paper is that selected directions within them need not be inert. All comparisons freeze the declared response and controls prospectively and match first-order response leakage; Section 4 gives the full protocol.

The evidence is organized around one central mechanism, two main results, one limited cross-architecture confirmation and one strict failure boundary. The central mechanism is that response-preserving freedom is a state-dependent distribution $\mathcal{V}_{\theta} = \ker DR(\theta)$, not a source-fixed null space. The functional premise for it — that selected response-fibre directions carry adaptive value under prospectively frozen, matched controls — is established in Section 3 in CNN, Tiny ViT, frozen ResNet-18 final-classifier and frozen GPT-2 LoRA-B settings, without implying that every kernel direction is useful.

The first main result concerns geometric consequences. In the declared CNN and ResNet constructions, current-kernel transport and source-fixed replay exhibit different finite realizability-cost scaling, while minute chart drift can be amplified by more than three orders of magnitude in active response residual.

The second main result concerns functional consequences on recorded optimizer paths. Under prospectively frozen low-response budgets, candidate scales are selected on calibration data and scored on disjoint held-out data. The instantaneous response kernel provides greater held-out learning utility than the recorded update and source-kernel, time-misaligned and signed-permuted controls.

This ordering is supported in 7/8 new seeds in a reduced ResNet-18/CIFAR-10 construction and confirmed in 8/8 new seeds and all 24 noninitial nodes in a GPT-2 rank-2 native-LoRA-B construction. The cross-architecture claim concerns the ordering of low-response-budget utility, not equality of response definitions, parameter dimensions or effect magnitudes.

The earlier GPTW-v3 audit addresses a separate geometric quantity: current-versus-source-fixed correction-cost scaling. It passes 8/8 new seeds and 32/32 interior nodes. These counts must not be identified with the 24-node Pareto confirmation. The two GPT-2 programmes are separately frozen protocols with different statistical objects, and their node counts are not interchangeable.

Two boundaries remain load-bearing. Recorded AdamW supplies real training nodes and update proposals, but ordinary AdamW is not shown to align naturally with the instantaneous fibre. Separately, the path minimizing the declared Moving-Fibre F16 action is not the least expensive path to realize. The results therefore support a local stability–plasticity role for moving response fibres without establishing optimizer naturality or a global variational law.

These are finite-chart mechanism claims with experiment-specific scopes. They do not establish ordinary-optimizer navigation, complete-kernel transport, full-parameter GPT-2 geometry or a global variational law. Any unifying variational functional must be derived independently rather than fitted to the observed ordering.

Section 2 defines the response-fibre geometry, and Section 3 establishes that selected response-preserving directions can carry adaptive value. Sections 4–6 test moving-kernel transport, finite realizability-cost scaling and transverse amplification. Section 6.5 tests the low-response learning frontier at recorded optimizer nodes in reduced ResNet and GPT-2/native-LoRA-B constructions. Sections 7–8 test and reject the proposed equivalence between restricted path action and realization cost. Sections 9–12 state the evidence ledger, interpretation, limitations and conclusions.

1.1 Relation to existing accounts

Five distinctions fix the object precisely.

Full-function symmetry versus declared response equivalence. Work on parameter-space symmetry [9], permutation re-basining [1] and linear mode connectivity [7] studies transformations that preserve the full function. A response fibre requires only $R(\theta') = R(\theta)$ on a prospectively declared anchor set, so it may contain genuine functional change invisible to that response. Exact function-preserving symmetries are a special subset; their adaptive value is not assumed.

Reparameterization invariance versus a state-dependent kernel. Flatness can be manipulated by reparameterization [4], so all declared parameter-space functionals require covariance checks and pulled-back metrics. The present claim is different: $\mathcal{V}_\theta$ is indexed by state, and the measured object is how this kernel changes along a realized path rather than how a fixed scalar changes under coordinates.

Image geometry versus kernel geometry. The neural tangent kernel [13] studies the image of a parameters-to-functions linearization. Here the object is the kernel of that linearization at a trained checkpoint within a declared parameter block. Natural-gradient and information-geometric methods [2] supply metric machinery but do not imply that selected null directions carry adaptive value.

Moving distributions versus measured adaptive value. That $\ker DR(\theta)$ varies with θ is standard geometry, not the empirical claim. The test is whether replacing a current response-kernel chart by a source-fixed chart creates a prospectively measurable realization penalty and whether small chart drift is amplified by the current response map.

Null-space protection versus response-fibre navigation. Orthogonal-update and gradient-projection methods [17, 6, 15] protect earlier computations; LoRA [11] constrains updates; task arithmetic [12] reuses learned differences. Here an external response is frozen prospectively, its kernel is recomputed along a path, and adaptive value, transport cost and path action are measured separately. The distinction is response-fibre navigation, not generic null-space protection.

2 Response fibres as functional freedom

Let $R(\theta)$ be an external response map, frozen prospectively on a declared anchor set before any evaluation. If y is a regular value, $R^{-1}(y)$ is a level-set submanifold. More generally, where R has constant rank in a neighbourhood of θ , the constant-rank theorem supplies the corresponding local level-set structure, and $T_\theta \mathcal{F}_y = \ker DR(\theta)$. At rank-changing points, only the linearized kernel is used here. At singular points no manifold structure is claimed, and $\ker DR(\theta)$ denotes only the linearized response-preserving space.

All audits below operate at trained checkpoints where the required Jacobian actions are evaluated by automatic differentiation, while numerical rank, kernel residuals and finite/JVP consistency are explicitly recorded; the full Jacobian need not be materialized, and the linearized object is well defined whether or not the level set is smooth there. The directions that leave the response unchanged to first order are the response-fibre tangents

$$DR(\theta)v = 0, \quad v \in \ker DR(\theta). \quad (2)$$

For a declared Riemannian or operational metric g on parameter space, the coordinate-covariant projection onto the fibre is

$$P_{\ker DR}^g = I - g^{-1}DR^\top (DR g^{-1}DR^\top)^\dagger DR. \quad (3)$$

The declared output norm is positive definite on the audited response coordinates; otherwise the operational zero set may strictly contain $\ker DR(\theta)$. The complementary directions are response-normal: motion along them changes the declared response and is subject to ordinary error correction. This split is the working definition of functional freedom. It is relative to R by construction, and nothing below claims an R -independent decomposition.

Terminology. Throughout this paper, behaviour means only the value of a prospectively declared finite response map R , not the model’s complete input-output function. Learning means improvement on the separately declared adaptation objective, evaluated on held-out data where specified. A response fibre therefore collects internal realizations that agree on the declared behaviour but may differ in their capacity for subsequent learning.

Two statements must be kept apart throughout. That a direction lies in $\ker DR$ is a structural fact, true of vast numbers of directions in any overparameterized model. That a direction carries adaptive value is a functional fact requiring norm-matched, leakage-matched controls. Section 3 addresses only the second.

3 Functional premise: adaptive value inside a frozen response fibre

This section establishes the functional premise for everything that follows: motion inside a response fibre is not inert redundancy. Throughout it the fibre is identified once, at a source checkpoint, and held fixed — an assumption Section 4 then removes.

3.1 Frozen response and matched controls

The audits in this section follow the common experimental logic below; the restricted Fisher-action audit in Section 7 follows its own separately frozen decision structure. The discipline is what makes the fibre claim testable at all.

1. **Prospective freezing.** The response map R , the anchor set, the internal representation Q , the metric, the dictionary basis where one is used, and all control arms are fixed before target labels or target gradients are observed. Held-out labels are evaluation-only.
2. **Admissibility before interpretation.** An audit enters its validity domain only if the source task was actually learned above a declared accuracy threshold and if the recorded trajectory can be replayed exactly. Stages failing this are reported as inadmissible and their mechanism outputs are treated as diagnostics, never as evidence (Section 9 and Supplementary Section SA).
3. **Norm and leakage matching.** Every compared arm carries the same tangent norm and retains its own realized response-normal component, and stepwise response-leakage mismatch is measured and gated. Without this, an apparent tangent advantage could simply be a larger step or a hidden response change. The fraction of each realized update that lies in $\ker DR$ is itself a protocol quantity and should be reported alongside any effect, since it bounds how much a tangent intervention can move at all.
4. **Structural controls, not straw men.** The primary control is a deterministic equal-norm random vector in the same frozen kernel. Additional arms include no-tangent motion, sign reversal, shuffled coefficients, random subspaces, normal-only motion, shuffled task descriptors, and development-mean substitutes.

Preregistered effect gates are declared in advance and a pooled significance test never overrides a failed magnitude gate. Pooled tests with ambiguous independent units are withheld; complete statistical decision rules are given in the Supplementary Protocol.

Unless stated otherwise, the models are a small spatial convolutional network (the CNN of the CNN–MNIST construction) and a tiny vision Transformer (Tiny ViT), the source domain is MNIST, and the independent target domain is USPS, an independently collected handwritten-digit dataset. The three declared systems of this paper are the CNN–MNIST construction, the ResNet-18/CIFAR-10 confirmation and the GPT-2 rank-2 LoRA-B subspace; after first use they are abbreviated CNN, ResNet and GPT-2/LoRA-B. TinyCNN/MNIST is reserved for the F16 and action-cost audits of Sections 7–8, where the exact three-parameter construction matters. Artificial label-bijection tasks share identical inputs and differ only in the target relation, which isolates task identity from input statistics.

3.2 Adaptive value of realized fibre motion

The core positive result: response-invisible directions are not inert redundancy. Earlier relaxation diagnostics suggested a normal–tangent timescale separation but did not close their causal gates; the complete result is retained in Supplementary Section SA. The load-bearing adaptive-value evidence is the controlled finite intervention below.

Learned tangents beat norm-matched random tangents under a synthetic shift. After source training the anchor Jacobian was frozen, and during adaptation the realized displacement was split into its normal and tangent components. The primary control kept the realized normal component and replaced only the learned tangent by a deterministic equal-norm random vector in the same kernel; a second control removed tangent motion. Parameter redundancy, tangent norm, and first-order response invisibility are thereby held constant across arms.

Architecture	True – random	True – no tangent	Seed wins	Leak mismatch
CNN	+12.96 pp	+11.94 pp	8/8 and 8/8	11.85%
Tiny ViT	+18.33 pp	+17.87 pp	8/8 and 8/8	11.47%

All frozen gates passed; across the sixteen paired comparisons the one-sided Wilcoxon result was $p = 1.5259 \times 10^{-5}$. Absolute source, pre-shift and per-arm accuracies are in Supplementary Table S6. Throughout this paper, difference columns report the median of paired per-seed differences, not the difference of the two medians; the two estimators need not coincide and the former is the quantity the paired tests are run on.

$\Delta\theta_{\text{learned}}^{\parallel}$ contributes more shifted-domain adaptation than an equal-norm random direction in $\ker DR$. (H2-F)

The advantage survives on an independently collected domain. The same comparison was run outside the synthetic shift: models trained on MNIST, adapted to USPS, response frozen on twelve MNIST anchors, and for each architecture and each of four seeds the learned tangent compared with eight deterministic equal-norm random kernel vectors and a no-tangent arm.

Architecture	True – random	True – no tangent	Consistent seeds	Median / p95 leak mismatch
CNN	+7.33 pp	+7.19 pp	4/4	17.44% / 19.40%
Tiny ViT	+8.99 pp	+8.86 pp	4/4	10.08% / 14.61%

All frozen gates passed and the pooled one-sided Wilcoxon p -value was 0.00390625. Absolute accuracies are in Supplementary Table S6.

$$A_{\text{USPS}}(\Delta\theta_{\text{learned}}^{\parallel}) > \text{median} A_{\text{USPS}}(\Delta\theta_{\text{random}}^{\parallel}) \quad (\text{H2-ID})$$

This establishes the functional premise used below. Section 6.5 tests the stronger pathwise question of whether the instantaneous response kernel provides superior held-out learning utility under a frozen finite response budget. The present comparison is conditional on the declared architectures, source anchors, optimizer, target adaptation set, and stepwise leakage gates. It does not show that every response fibre carries semantic capacity, that the result survives natural distribution shift or large models, or that the learned direction minimizes any variational functional. This frozen-fibre adaptive-value audit is a functional premise at a fixed chart; it is not the real-node low-response frontier of Section 6.5, and the two must not be pooled.

3.3 Task conditioning and limits

Three frozen label-bijection tasks used identical inputs but distinct target relations. Each target retained its own response-normal sequence while tangent content was exchanged across tasks, with exactly matched tangent norms and median leakage mismatch below 12%.

Quantity	CNN	Tiny ViT
Task diagonal advantage	+34.39 pp	+21.63 pp
Own tangent – random	+32.44 pp	+19.53 pp
Own tangent – sign reversal	+45.89 pp	+24.86 pp
Diagonal winners	18/18	18/18
Correct-gradient Spearman	0.671	0.359
Shuffled-gradient Spearman	−0.167	−0.006

The total effect was large and fully consistent, but the complete mediation claim did not pass: the Tiny ViT correct-gradient correlation fell below the frozen 0.50 gate. Permitted: tangent value depends on task identity and direction sign. Prohibited: raw Euclidean gradient alignment universally selects the tangent.

A separate audit froze a correction learned on affine label bijections and evaluated it on an independently specified non-affine family under identical source-local geometry, replay state and evaluation budget. Useful structure survived transfer, so the correction is not mere within-family memorization; but the frozen cosine and magnitude gates failed in both architectures, and the cross-family shared-correction claim is not supported as stated. The transferred object is better described as a coarse, architecture-dependent prior than as a universal functional direction. The shared-correction tables, coefficient cosines, pooled orderings and per-gate outcomes are in Supplementary Tables S4–S6.

3.4 Internal premise confirmation: frozen ResNet-18 representation

Frozen scope. A torchvision ResNet-18 with ImageNet-1K pretrained weights, backbone entirely frozen, on CIFAR-10. The response and the fibre are declared on the complete 5,130-parameter final classifier, so no subchart approximation enters. Sixteen new seeds, 64726–64741, at 4, 16 and 32 anchors, with 16 anchors as the primary gate; 32 matched random kernel directions per seed; confirmation labels used for evaluation only; float64 Jacobian, SVD and finite-step verification.

Frozen gates. At least 12/16 seeds at the primary anchor count and at least 10/16 at the sensitivity counts, with per-seed thresholds $\Delta L_{\text{true}} \leq -0.001$, $\Delta L_{\text{anti}} - \Delta L_{\text{true}} \geq 0.002$, $\Delta L_{\text{random,best}} - \Delta L_{\text{true}} \geq 0.001$ and $\Delta L_{\text{shuffled}} - \Delta L_{\text{true}} \geq 0.001$. Numerical eligibility required $\|Jv\|_2 / (\|J\|_2 \|v\|_2) \leq 10^{-12}$ and $\|\Delta R\|_\infty \leq 10^{-9}$.

Quantity	4 anchors	16 anchors	32 anchors
Candidate seeds	16/16	16/16	16/16
Projected-gradient share, median	0.453	0.338	0.282
True loss change, median	−0.02705	−0.01842	−0.01450
Anti-true margin, median	0.06428	0.04149	0.03162
Best-random-true margin, median	0.02616	0.01784	0.01380
Shuffled-true margin, median	0.02624	0.01742	0.01374
Max kernel residual	3.70e−14	2.27e−13	4.22e−13
Max finite response leakage	5.51e−14	2.98e−13	5.06e−13

Every seed passed every frozen gate at all three anchor counts. Increasing the response declaration reduced the median projected-gradient share from 0.453 to 0.282 and reduced, but did not eliminate, the adaptive advantage. The result therefore confirms the ordering of selected response-fibre tangents over matched controls across anchor-set size; it does not establish anchor-independent effect magnitude.

The backbone was not adapted and the response kernel was not transported. This confirms selected tangent value outside the original construction; it does not replicate the moving-fibre or realizability-cost results of Sections 4–8.

3.5 GPT-2 external-confirmation programme: frozen-fibre premise

Frozen scope. A frozen GPT-2 backbone with rank-2 LoRA adapters. The response and the fibre are declared on the LoRA-B parameters of the final two transformer layers, on built-in codeword prompt tasks. Eight new seeds; sixteen matched random kernel directions per seed; confirmation labels used for evaluation only.

Result. In this frozen-fibre codeword-prompt protocol, the selected response-kernel tangent outperformed the no-tangent, sign-reversed, shuffled-target and best-of-sixteen random-kernel controls in 8/8 seeds of that protocol. Every frozen functional, effect-size and response-preservation gate passed. These eight seeds belong to the frozen-fibre premise audit and are distinct from the eight seeds of the GPTW-v3 correction-cost audit of Section 5.4 and from the eight seeds of the low-response Pareto audit of Section 6.5.3.

This audit extends the adaptive-value premise across modality. It concerns a rank-2 LoRA-B block in the last two layers under built-in codeword prompts, not the complete GPT-2 parameter space or general language understanding. This Section 3.5 audit freezes the kernel and tests no moving path. Section 5.4 reports both a constructed current-versus-fixed codeword audit and its prospective confirmation on disjoint natural-English continuations. Together, the Section 3 results establish only the functional premise for what follows: selected response-fibre motion need not be inert.

4 Central mechanism: causal navigation of moving response fibres

4.1 From a source kernel to a moving distribution

Every audit in Section 3 freezes $\mathcal{V}_\theta = \ker DR(\theta)$ at a source checkpoint and works inside it. The construction is legitimate, and Section 2 states it as a first-order object at a point. But nothing in that construction says the fibre stays put. As parameters move, $DR(\theta)$ changes, and with it the space of response-preserving directions. Two readings are then possible and they are not equivalent.

Static reading. The response fibre is treated as one source-fixed subspace. An admissible response-preserving intervention is generated inside it, and identifying it at the source point is assumed to lose nothing over the distances travelled.

Moving reading. The response fibre is a distribution $\mathcal{V}_\theta$ varying with θ . Motion that is response-preserving at the source becomes progressively less so as the path advances, and a stale fibre is a decreasingly good description of what the network is actually free to do.

Navigation is used operationally throughout, not rhetorically: it means generating each admissible tangent proposal from current and past information while maintaining membership in the current response kernel. Writing H_{static} for $\mathcal{V}_{\theta_k} \approx \mathcal{V}_{\theta_0}$ and H_{moving} for $\mathcal{V}_{\theta_k} \neq \mathcal{V}_{\theta_0}$, the two readings make a testable prediction that differs in kind, not in degree. Under the static reading, replaying a trajectory against the source fibre and against the current fibre should cost the same up to noise. Under the moving reading, the two should diverge systematically with distance travelled, and the divergence should have a measurable scaling.

4.2 Causal online navigation protocol

Both arms begin from the same trained source model and receive the same ordered adaptation minibatches, response anchors, metric probes, capability endpoint, Fisher step radius, Newton retraction rule and computational budget. They differ only in the chart used to generate each tangent proposal. They are therefore two online paths matched in source state, data order, response target, step-radius schedule,

capability endpoint and budget — not two realizations of one parameter displacement, and not guaranteed to share a parameter endpoint or a total displacement.

Moving arm. The chart is transported and re-projected into the current kernel at each step,

$$B_k = \text{TransportProject}(B_{k-1}, \ker DR(\theta_k)), \quad \|DR(\theta_k)B_k\| \leq \epsilon_{\ker}, \quad (4)$$

with $B_k \in \mathbb{R}^{N \times 8}$. The construction is causal: it uses current and past information only, and reads no future minibatch, no future Jacobian and no final result. It is a current-kernel transported chart, not a dynamically solved 4146-dimensional kernel.

Fixed arm. $B_k = B_0$ throughout, but every step still passes through the same Newton response retraction. The fixed arm is therefore not a deliberately inadmissible straw man: it is a stale tangent space combined with an identical constraint-maintenance mechanism.

Realizability cost. A trial step is taken in the chart and then retracted onto the frozen response,

$$\theta_{k+1}^{\text{trial}} = \theta_k + B_k \Delta z_k, \quad \theta_{k+1} = \text{Retr}_{R(\theta_0)}(\theta_{k+1}^{\text{trial}}), \quad (5)$$

the retraction being a Newton response correction. The single-step realizability cost is the local output divergence the retraction had to remove, and the path cost is its sum:

$$c_k = \sqrt{2D_{\text{KL}}(p_{\theta_{k+1}^{\text{trial}}} \| p_{\theta_{k+1}})}, \quad C(h) = \sum_k c_k. \quad (6)$$

C measures the local output deviation that must be removed in order to turn a proposed tangent step into an actual response-preserving one; a chart that is exactly right incurs none of it. The divergence is measured on the frozen metric probes, so C is a retraction Fisher-length proxy — not a physical energy, a wall-clock duration, or the only possible notion of realizability cost.

Step radii and fitting. Each arm is run at four step sizes, $h \in \{0.08, 0.04, 0.02, 0.01\}$, and a single exponent is fitted per seed by ordinary least squares on

$$\log C(h) = \alpha \log h + \beta. \quad (7)$$

Exponents are reported as medians over comparable seeds, with fit quality alongside. The moving arm predicts $\alpha_{\text{moving}} > 0$; the fixed arm predicts $\alpha_{\text{fixed}} \approx 0$.

5 Main result I: realizability-cost scaling

5.1 Frozen hypotheses and gates

Frozen gates. Eligibility: source accuracy ≥ 0.90 ; source-chart kernel residual $\leq 10^{-7}$; Fisher metric certification passed; all eight paths admissible; at least 14/16 seeds fully comparable. Per path: response-retraction relative error ≤ 0.002 ; retraction-to-step ratio ≤ 0.20 ; moving-kernel residual $\leq 10^{-5}$; effective metric rank ≥ 8 ; maximum step principal angle ≤ 0.80 rad; action left-versus-trapezoid relative change ≤ 0.08 . Scaling: $\alpha_{\text{moving}} \geq 0.50$ with $R_{\text{moving}}^2 \geq 0.80$; $|\alpha_{\text{fixed}}| \leq 0.25$ with relative span $(\max C_{\text{fixed}} - \min C_{\text{fixed}}) / \text{median } C_{\text{fixed}} \leq 0.20$; separation $\alpha_{\text{moving}} - \alpha_{\text{fixed}} \geq 0.25$; cost ratio $C_{\text{fixed}} / C_{\text{moving}} \geq 2$ and residual ratio $r_{\text{fixed}} / r_{\text{moving}} \geq 2$. At least 12 seeds must pass every substantive gate. Full per-gate outcomes are in the supplementary gate table.

One seed was excluded: seed 70741, whose source accuracy was 0.898 against the frozen 0.900. Its metric and chart certifications passed; it failed source-model eligibility and was excluded under the frozen rule before any comparison.

5.2 Prospective 16-seed confirmation

Fifteen of sixteen seeds retained full comparability, and all fifteen passed every preregistered gate.

Quantity	Value	Unit
Fully comparable seeds	15/16	count
Seeds passing all frozen gates	15/15	count
Median moving exponent α_{moving}	0.684	dimensionless
Moving exponent range	0.583–0.863	dimensionless
Median fixed exponent α_{fixed}	0.00715	dimensionless
Fixed exponent range	−0.0124–0.0489	dimensionless
Median moving R^2	0.995	dimensionless
Minimum moving R^2	0.976	dimensionless
Median fixed relative span	3.05%	relative
Maximum fixed relative span	11.16%	relative
Median cost ratio	21.30	dimensionless
Smallest-radius cost-ratio range	13.61–26.17	dimensionless
Tangent residual ratio	21.47	dimensionless
Excluded seed 70741 source acc.	$0.898 < 0.900$	—

The separation is not carried by a few outlying seeds. The moving exponent stays between 0.583 and 0.863 with median $R^2 = 0.995$ and minimum 0.976, so the power law is well determined in every comparable seed; the fixed-arm exponent was close to zero under the prospectively frozen absolute-exponent and relative-span gates, with median 0.00715 and range −0.0124 to 0.0489, and a median relative span of 3.05% across the four radii. The cost ratio at the smallest radius ranges from 13.61 to 26.17.

Post-confirmation leave-one-radius-out refits, which were not a preregistered gate, preserved the frozen slope-separation gate; complete per-seed and refit records are archived.

The two ratios agree closely with one another (21.30 and 21.47). Their agreement is consistent with a common geometric source, but does not identify that source or establish a causal relation with fibre rotation.

5.3 Cross-model confirmation in ResNet-18 / CIFAR-10

The cost-scaling separation is not confined to the original construction. A prospective sixteen-seed audit used a frozen ImageNet-pretrained ResNet-18 backbone with a trainable terminal residual adapter and classifier, and applied the same current-kernel transport and source-fixed replay comparison.

Quantity	Value
Seeds passing every geometric and held-out gate	16/16
Moving-minus-fixed slope separation, median	0.575
Smallest-radius fixed / moving cost ratio, median	2.997
Raw development / confirmation gradient cosine	0.414
Projected cosine	0.996

The permitted conclusion is that the realizability-cost scaling difference between current-kernel transport and source-fixed replay is not restricted to the CNN–MNIST construction. It is not a universal scaling law: the result remains specific to this ResNet construction, these finite radii and this transported chart.

5.4 GPT-2 confirmation of current-versus-fixed correction-cost scaling

This subsection reports a correction-cost audit and is distinct from the low-response Pareto audit of Section 6.5. Its 32 interior nodes are not the 24 noninitial nodes used in the Pareto confirmation, and its seeds, response map and statistic are separately frozen.

An analogous current-kernel versus source-fixed comparison in frozen GPT-2 rank-2 LoRA-B (the GPTW-v3 correction-cost protocol) first passed 6/6 seeds and 18/18 codeword-prompt nodes, then passed a prospective confirmation on disjoint frozen natural-English development and response sets in 8/8 new seeds of that protocol and 32/32 interior nodes. At state ϕ , with $J_\phi = DR(\phi)$, response error $e_h^{(a)} = R(\phi + hv^{(a)}) - R(\phi)$ was assigned the regularized Newton correction

$$\delta\phi_{N,h}^{(a)} = -J_\phi^\top (J_\phi J_\phi^\top + \lambda I)^{-1} e_h^{(a)}, \quad C_{\text{GPT}}^{(a)}(h) = \|\delta\phi_{N,h}^{(a)}\|_2. \quad (8)$$

C_{GPT} is a correction norm in the declared LoRA-B coordinates, not the CNN Fisher-length proxy.

The approximately quadratic current-arm exponent and approximately linear fixed-arm exponent are analytically induced by the arm definitions and are reported as numerical-correctness checks, not as independent evidence of moving-fibre geometry. The non-trivial content of this audit is threefold: the finest-radius fixed-to-current correction-cost ratio of 8.87–35.56, which is not fixed by the expansion; the measured fibre rotation between the current and source charts along the path (principal angle 0.0161–0.0713 rad in the codeword v2 stage); and the response-path and retraction certification, with maximum post-retraction response error $1.08\text{e}-15$.

Quantity	Range
Seeds / interior nodes passing	8/8; 32/32
$\alpha_{\text{current}} / \alpha_{\text{fixed}}$	1.9987–2.0040 / 1.0625–1.2106
Finest-radius $C_{\text{fixed}}/C_{\text{current}}$	8.87–35.56
Current-minus-fixed slope gap	0.792–0.939
Active-Jacobian residual ratio	1.41e7–1.12e8
Max post-retraction response error	1.08e–15

Because re-projection eliminates the current arm’s first-order response, Taylor expansion forces its correction cost to $O(h^2)$; a surviving fixed-arm linear term forces $O(h)$. Thus the current exponent near 2, fixed exponent near 1 and their slope gap are analytic-order numerical-correctness checks, not independent evidence of fibre motion. The non-trivial finite effects are magnitudes not fixed by this expansion: in codeword v2, the finest-radius cost ratio is 8.60–25.70, the principal angle is 0.0161–0.0713 rad and the path response error is approximately $6.7\text{e}-16$; in V3 natural text, the cost ratio is 8.87–35.56 and the active-Jacobian residual ratio is $1.41\text{e}7$ – $1.12\text{e}8$. These results satisfy the frozen controls but do not establish semantic invariance, ordinary training navigation or a global variational law.

6 Main result II: transverse amplification in a trainable ResNet block

A moving fibre raises a question the cost measurements do not answer: how far must the fibre turn before the difference matters? The answer is that it need not turn far at all.

6.1 Active directions

At a point on the path, let B_m be the transported and re-projected current chart and B_0 the source-fixed chart. Both have Euclidean-orthonormal columns after transport and re-projection, so $B_m B_m^\top$ and $B_0 B_0^\top$ are the corresponding chart projectors. The two descent proposals inside them are

$$d_m = -\frac{B_m B_m^\top \nabla L}{\|B_m B_m^\top \nabla L\|}, \quad d_f = -\frac{B_0 B_0^\top \nabla L}{\|B_0 B_0^\top \nabla L\|}, \quad \delta d = d_f - d_m. \quad (9)$$

6.2 Transverse amplification

Here $J = DR(\theta_k)$ is the Jacobian of the prospectively frozen response map evaluated at the current path state. The response declaration is frozen; the Jacobian value is not — this is exactly what separates $DR(\theta_k)$

from $DR(\theta_0)$. Two distinct ratios must be kept apart. The active residual ratio compares how visible the two proposals are to the frozen response,

$$\|Jd_f\|/\|Jd_m\|, \quad (10)$$

while the transverse gain contrast compares the response gain along the difference direction with the gain along the current-fibre direction,

$$A_{\perp} = \frac{\|J\delta d\|}{\|\delta d\|}, \quad \Gamma_{\perp} = \frac{\|J\delta d\|/\|\delta d\|}{\|Jd_m\|/\|d_m\|}. \quad (11)$$

These are not the same quantity and are never reported as one.

6.3 Prospective confirmation

A sixteen-seed prospective audit on a trainable ResNet-18 layer4-plus-classifier system, using an eight-dimensional matrix-free transported response-kernel chart, passed with 16/16 against a frozen 12/16 gate.

Quantity	Median
Seeds passing (frozen gate 12/16)	16/16
Direction cosine between d_m and d_f	0.999990
Active residual ratio $\ Jd_f\ /\ Jd_m\ $	1437.844
Transverse response gain $A_{\perp}$	90.108
Transverse gain contrast $\Gamma_{\perp}$	341,763
Finest finite-difference / JVP ratio	0.999998
JVP additivity error	5.05e−14

Two parameter directions that are almost collinear need not be functionally equivalent in the current response geometry: a minute angular drift of the fibre can fall into a high-gain transverse direction. The finite-difference agreement shows this is a property of the response map, not a linearization artifact.

6.4 Precision certification and its failure history

The signal gates of the first attempt passed, but its JVP identity certification failed under float32 with TF32 matrix multiplication enabled. The repair ran in float64 with TF32 disabled and passed development certification; the confirmation above then used a fresh set of sixteen seeds. The precision failure was caught by a preregistered certification gate, and no result from the failed stage is reported as positive.

6.5 Functional consequence: low-response learning freedom

The geometric results above separate the instantaneous response kernel from stale charts. We next ask whether that distinction has a held-out learning consequence at real optimizer nodes. The recorded AdamW path supplies the nodes and the update proposals only; the successful instantaneous-kernel arm is a counterfactual projection of those recorded updates into the current kernel. Ordinary optimizer navigation is not established by this audit and no arm here is evidence that AdamW moves along the fibre of its own accord.

6.5.1 Frozen causal question

For arm a and scale α , let $C_a(\alpha)$ be the norm of the change in the prospectively declared response. For each frozen budget $\varepsilon \in \{0.02, 0.05, 0.10, 0.20\}$, calibration data select the scale that maximizes loss reduction subject to $C_a(\alpha) \leq \varepsilon$. The selected scale is then scored once on disjoint held-out data; no held-out quantity enters the selection. The primary statistic is the area under this four-budget utility frontier (AUC). Controls

are the recorded optimizer update, the source kernel, a half-path time-shifted kernel, and a deterministic signed permutation. Every arm, including the controls, is selected and scored under the same four frozen budgets, the same finite α grid and the same calibration/held-out split, so no arm receives a wider search than another.

Important interpretive constraint. Low first-order response cost is partly definitional for a kernel direction, and by itself carries no evidential weight here. The nontrivial evidence is threefold: that the calibration-selected finite steps retain held-out learning utility at all; that a scale chosen on calibration data generalizes to the disjoint held-out set; and that the instantaneous kernel systematically beats the stale source chart, the time-shifted chart, the signed-permuted chart and the recorded unprojected update under the same frozen finite response budgets.

6.5.2 Reduced ResNet: supported ordering

The reduced CPU construction uses frozen ImageNet ResNet-18 features with a trainable adapter and head. Seven of eight prospectively new seeds pass the complete frozen ordering, exceeding the six-seed decision threshold. The ordering is therefore supported, not confirmed, in this construction.

ResNet audit	Observed	Decision role
Supporting seeds	7/8	primary gate
Current positive	51/56 nodes	held-out utility
Current > source	51/56 nodes	stale-chart control
Current > far	49/56 nodes	time-shift control
Current > permuted	52/56 nodes	structure control

Pooled median AUC: current 0.000790; source 0.000681; half-path time-shifted 0.000728. This is supporting rather than universal evidence because one seed fails the full ordering, and the reduced CPU construction is not the full training configuration.

6.5.3 GPT-2 native-LoRA-B confirmation

The cross-architecture audit freezes the standard GPT-2 backbone and trains rank-2 native LoRA-B coordinates in layers 10 and 11 (9,216 trainable parameters). Candidate scales are selected on calibration prompts and evaluated on disjoint natural-language templates. All 8/8 new seeds of this low-response Pareto protocol and all 24/24 noninitial path nodes pass the prospectively frozen ordering, which is confirmation within the declared protocol only. These seed and node counts belong to the Pareto audit and are not those of the GPTW-v3 correction-cost audit of Section 5.4.

GPT-2 audit	Observed	Pooled median AUC
Supporting seeds	8/8	—
Current kernel	24/24 positive	0.11858
Source kernel	current wins 24/24	0.03341
Half-path shifted	current wins 24/24	0.03641
Recorded AdamW	current wins 24/24	0.00851
Signed permutation	current wins 24/24	0.00853

6.5.4 Claim boundary

The ResNet and GPT-2 audits share a decision structure, not an identical response map, parameter dimension or effect scale. The GPT-2 task uses an artificial color-code mapping expressed through disjoint natural-language templates; it is not a claim of general semantic invariance. The Pareto frontier is evaluated on a finite α grid and four discrete budgets, not optimized continuously. The random frozen LoRA-A factor is part of the seed axis. Most importantly, projection of recorded updates

into the instantaneous kernel is a counterfactual intervention: ordinary optimizer navigation is not established. These results show locally usable stability–plasticity freedom only. The two low-response result packages are not yet contained in the cited public repository snapshot and are recorded as such in the availability statement.

Updated hierarchy. One central mechanism: $\mathcal{V}_\theta = \ker DR(\theta)$ moves with state. Two main results: moving geometry has finite, transversely amplified consequences; and its instantaneous kernel yields a superior held-out learning frontier under frozen response budgets. GPT-2/native-LoRA-B supplies the strongest cross-architecture confirmation of the second result, not the theoretical centre of the paper. Optimizer naturality and a global variational law remain strictly outside the claim.

7 Restricted variational test: Moving-Fibre F16 ordering

Once the fibre is known to move, a natural follow-up is whether some functional selects a path within it. The direction of inference matters: the moving-fibre geometry of Sections 4–5 suggests a candidate path-selection functional, not the reverse.

7.1 Frozen question

The question is whether the natural-gradient path minimizes a declared action among a fixed set $\mathcal{A}_6$ of six causal online algorithms, at every tested step radius:

$$\gamma_{\text{nat}} \stackrel{?}{\in} \arg \min_{\gamma \in \mathcal{A}_6} S_{\text{MF16}}[\gamma; h], \quad h \in \{0.08, 0.04, 0.02, 0.01\}. \quad (12)$$

The action is a discrete functional on the realized sequence. At step k , with transported chart B_k , let $G_k = \text{regularize}(B_k^\top F_{\theta_k} B_k)$ be the eigenvalue-floored output-Fisher tensor in chart coordinates and let $g_k = B_k^\top \nabla L$ be the capability gradient in the same coordinates. The capacity factor is

$$H_k = h_0 + \frac{\sqrt{g_k^\top G_k^{-1} g_k}}{s_0}, \quad h_0 = 0.1, \quad (13)$$

with s_0 the same quantity evaluated at the source point, so that H_k is dimensionless and equals $h_0 + 1$ at the source. For the realized increment $\Delta\theta_k = \theta_{k+1} - \theta_k$, write its chart coordinates at the two endpoints as $z_k^- = B_k^\top \Delta\theta_k$ and $z_k^+ = B_{k+1}^\top \Delta\theta_k$, and the corresponding Fisher lengths as $d\ell_k^- = \sqrt{z_k^{-\top} G_k z_k^-}$ and $d\ell_k^+ = \sqrt{z_k^{+\top} G_{k+1} z_k^+}$. The action is the trapezoidal sum

$$S_{\text{MF16}}[\gamma; h] = \sum_{k=0}^{K-1} \frac{1}{2} \left(\frac{d\ell_k^-}{H_k} + \frac{d\ell_k^+}{H_{k+1}} \right). \quad (14)$$

Both the metric and the capacity are therefore re-evaluated at every node, and each increment is measured in the chart at both of its endpoints. S_{MF16} carries the units of Fisher length; H is dimensionless by the source normalization. The step radius h is the algorithmic Fisher step radius, not an integration mesh: the nodes are the realized steps, and K is set by first arrival at the capability endpoint. The pure Fisher length used below is the same sum without the H divisors. The frozen action-convergence statistic, the wrong-metric arm and the serialized implementation are specified in Supplementary Item 13.

Every algorithm is generated online and causally; recomputes the current response kernel at each step; uses the transported eight-dimensional chart of Section 4.2; reaches the common 20% capability-loss-reduction endpoint; and shares the data order, metric protocol and retraction rule. The construction is therefore the same moving-current-kernel construction as Section 5, applied to a path-selection rather than a cost-scaling question.

7.2 Frozen confirmation gates

The core preregistered gates are: at least 14/16 seeds fully comparable; natural gradient winning at all four radii in at least 12/16; beating the wrong-Fisher arm in at least 12/16; positive cost scaling for all six algorithms; fine-radius action convergence for all six; and natural gradient winning under the wrong metric in at most 2/16. Full thresholds are in the supplementary gate table.

7.3 Prospective result

Fourteen of sixteen seeds retained full comparability, and all fourteen eligible seeds passed. In every one of the fourteen, true natural gradient attained the lowest Moving-Fibre F16 action at every tested radius.

Quantity	Value
Seeds attempted / fully comparable	16 / 14
Natural gradient is the action minimizer at all four radii	14/14
Natural gradient beats the wrong-Fisher arm	14/14
Natural gradient wins under the wrong metric	0/14
Median natural cost exponent $\tilde{\alpha}_{\text{nat}}$	0.716
Median fit quality $\tilde{R}^2$	0.993
Median action advantage over the best competing algorithm ($h = 0.08$)	22.8%
Same, seed range	6.3%–41.1%

The same winner is obtained from pure moving-Fisher length, so the present experiment confirms the restricted ordering but does not isolate an independent contribution from the capacity normalization.

The conclusion is an argmin statement over the frozen six-algorithm family $\mathcal{A}_6$ only; no claim is made about minimization over arbitrary causal paths.

Seeds 73730 and 73732 were excluded because the coarse-radius natural arm exceeded the frozen `wrong_left_trapezoid_relative_change` threshold, at 0.10676 and 0.10358 against 0.08. Their source, metric, kernel, endpoint and response-retraction diagnostics otherwise remained regular; the frozen exclusion was retained without post-hoc relaxation. The result is therefore 14/16 eligible, 14/14 eligible seeds passed — not 14/14 of the seeds attempted.

8 Strict failure boundary: action minimization is not realization-cost minimization

The strongest counterintuitive result is not merely that the response fibre moves, but that the functional selecting a path and the cost of realizing that path produce systematically different orderings.

Section 5 measures what it costs to realize a proposed step; Section 7 measures which path minimizes a declared action. If the action were simply tracking realizability, the same arm would win both. It does not:

$$\gamma_{\text{nat}} \in \arg \min_{\gamma \in \mathcal{A}_6} S_{\text{MF16}}[\gamma] \quad \text{while} \quad \gamma_{\text{nat}} \notin \arg \min_{\gamma \in \mathcal{A}_6} C_{\text{real}}[\gamma]. \quad (15)$$

The gap is not marginal. At the smallest radius,

$$C_{\text{real}}^{\text{nat}}(h_{\min}) / \min_a C_{\text{real}}^a(h_{\min}) = 2.25 \text{ (median),} \quad \text{range } 1.60\text{--}4.19, \quad (16)$$

and the ratio stayed above 1.5 in all fourteen comparable seeds. The action-minimizing path is systematically more expensive to realize than the cheapest admissible alternative, by a factor that never falls to unity.

The separation does not come from the capacity normalization. A natural objection is that both compared quantities are declared here, so their disagreement could be an artifact of the capacity factor H in Equation (13). It is not. Removing H entirely and ranking the arms by pure moving-Fisher length gives the same winner as the action in 14/14 comparable seeds, and that winner is again natural gradient in 14/14. The separation therefore holds between the moving-Fisher length — a standard information-geometric object that this paper does not define — and the retraction cost. It does not depend on h_0 , on the source normalization, or on the capability probe.

This is a result, not a defect of either measurement. The safe form of the claim keeps every qualifier: the experiment rejects the identification of the declared restricted path action with raw realizability cost in the audited system — six declared algorithms, four frozen radii, one architecture and dataset. It does not show that path-selection functionals differ from costs in general, that optimizers actively pay a premium, or that any two-objective principle governs neural training. Nor does it contradict analytical mechanics, where an action was never required to equal an energy or a cost; what is falsified is an order-preserving identification that is easy to assume in a learning setting.

The open question is therefore not only whether a functional is globally minimized, but why the lowest-action path is willing to pay a higher realization correction.

9 Evidence ledger: negative results and claim boundaries

The table lists the load-bearing claims only. Untested and unsupported items — complete response-kernel transport, full-backbone transport, the residual-as-address and universal-correction hypotheses, a smooth global bundle, and global minimality under ordinary learning — are stated in Section 11 and recorded with their diagnostics in Supplementary Section SA. Frozen exclusions and their diagnostics are reported in Sections 5 and 7 and in Supplementary Section SA.

Claim	Status	Decisive evidence	Boundary
Selected response-fibre directions carry adaptive value	Confirmed across declared constructions	CNN / Tiny ViT controlled interventions on an independent domain, 8/8 runs, leakage matched; ResNet 16/16 external confirmation at 4, 16 and 32 anchors against sign-reversed, shuffled-target and best-of-32 random-kernel controls	Finite CNN / Tiny ViT audits and the frozen-classifier ResNet representation
Current-versus-fixed correction-cost scaling in GPT-2/LoRA-B	Supported	GPTW-v1 established selected tangent value in 8/8 seeds of the frozen-fibre premise protocol; V3 confirmed current-versus-fixed scaling on 8/8 seeds and 32/32 natural-text nodes, with α_{current} 1.9987–2.0040 against α_{fixed} 1.0625–1.2106 and cost ratio 8.87–35.56	Rank-2 LoRA-B of the last two layers; disjoint frozen natural-English sets; current direction re-projected at every node; correction-cost statistic only, distinct from the Pareto audit
Moving-fibre cost scaling	Confirmed in CNN and ResNet constructions	CNN–MNIST 15/16 eligible, 15/15 passed, $\alpha = 0.684$ against 0.007, cost ratio 21.30; ResNet-18/CIFAR-10 16/16, slope separation 0.575, cost ratio 2.997	Finite charts and radii; not a universal scaling law
Transverse response amplification	Confirmed	16/16 against a frozen 12/16 gate; direction cosine 0.999990 with active residual ratio 1437.8 and gain contrast 3.4e5; finite/JVP ratio 0.999998	Trainable layer4 + fc, eight-dimensional chart
Instantaneous response-kernel directions define a superior low-response held-out learning frontier at recorded optimizer nodes	Supported in reduced ResNet; prospectively confirmed within the declared GPT-2/native-LoRA-B protocol	Reduced ResNet v4.6: 7/8 new seeds. GPT-2/native-LoRA-B Pareto: 8/8 new seeds and 24/24 noninitial nodes, pooled median AUC 0.11858 for the current kernel against 0.03341 (source), 0.03641 (half-path shifted), 0.00851 (recorded AdamW) and 0.00853 (signed permutation)	The arm is a counterfactual kernel projection; ordinary optimizer alignment is not established. Budgets and scale grids are discrete. The two architectures use different response maps and parameter charts
Restricted Moving-Fibre F16 ordering	Confirmed	14/16 eligible; natural gradient wins at all four radii in 14/14; wrong metric 0/14	Six algorithms, CNN–MNIST
Restricted action and realizability cost induce the same path ordering	Refuted	Natural-gradient cost ratio median 2.25, above 1.5 in 14/14; the same holds for pure moving-Fisher length, so the separation is not produced by the capacity factor	TinyCNN/MNIST, $\mathcal{A}_6$; no combined scalar law is determined

Full failed, excluded and diagnostically ineligible stages — the early cross-architecture attempts, the developmental and diagnostic Fisher cohorts, statewide dictionary transport, the temporal-selection and local-variation failures, and the GPT-2 distance audits — are retained with their values in Supplementary Section SA. Inadmissibility is evidence neither against a hypothesis nor for it.

10 Discussion: from parameter storage to fibre navigation

Three layers of “navigation” must be kept apart. Confirmed: an online transported chart can use current and past information to maintain $\text{im } B_k \subseteq \ker DR(\theta_k)$ throughout a path. Supported as interpretation: functional

freedom is better represented as a state-dependent family than as a source-fixed null space. Not shown: that ordinary SGD, Adam or large-model training generally follows a globally minimal action. Navigation here is an operational property of causally generated current-kernel paths, not a theorem that unconstrained neural training naturally follows a horizontal lift or a globally minimal trajectory.

The Pareto result changes the role of the response fibre. The fibre is not only a set of first-order response-invisible directions whose chart moves with state. Within the declared finite charts, its instantaneous kernel also defines a measurable stability–plasticity frontier: more held-out learning can be obtained before the frozen response budget is exhausted. This is a functional consequence of using the current chart rather than a claim that every kernel direction is useful.

Recorded AdamW provides real training nodes and proposals, but the successful arm is produced by counterfactual projection. The experiment therefore does not show that AdamW naturally follows the response fibre. In this paper, ordinary optimizer navigation is not established.

The evidence ladder is short: selected null directions can carry adaptive value, the tested response-preserving chart moves, stale replay is costly, minute transverse departures can be strongly amplified, and the current chart yields a better low-response learning frontier than stale, time-shifted, permuted or unprojected alternatives; the per-layer judgment table is in the Supplementary Information.

The evidence supports a third view of neural adaptation: relevant structure resides neither solely in individual parameters nor solely in the update rule, but in a state-dependent family of response-preserving directions used by the declared counterfactual interventions at recorded learning nodes. This reframes overparameterization without treating the entire kernel as functional: random controls show that adaptive value is concentrated in a comparatively small, geometrically identified part of a large null space, and identifying that part prospectively remains open.

10.1 Variational outlook: a constrained open problem

The confirmed result is a moving-fibre mechanism, not a global variational law. A global principle would quantify over a class of causal, realizable paths, but Sections 4–5 show that this class cannot be represented by a fixed source kernel: realizability depends on a response fibre that changes with state. The restricted F16 audit adds a second constraint. Moving-Fisher action and Newton-retraction cost select different paths, so neither quantity can be substituted for the other or combined with a fitted coefficient after the fact.

Any continuum theory must therefore derive its admissible path class and its objective independently. Plausible directions include a genuinely multi-objective Pareto formulation, a renormalized realization-cost density, or a higher-order term controlling connection, transverse sensitivity, or curvature. The amplification result makes the last possibility concrete: a small angular drift can enter a high-gain response-normal direction even when the two parameter directions remain nearly collinear. No such term is inserted into the present action, and no coefficient is inferred from the observed ordering.

The order of inference matters. Moving-fibre geometry determines which online proposals are response-preserving at the current state; only after that realizable class is specified does global path variation become meaningful. The present data therefore constrain future theory without repairing the declared F16 action after the fact.

11 Limitations

11.1 Model scope

Each confirmed claim carries its own model scope, and they are not interchangeable. Selected tangent value is confirmed in the complete final-classifier parameter space of a frozen ImageNet-pretrained ResNet-18 representation, across three anchor counts and sixteen seeds, in addition to the original CNN and ViT audits. Moving-fibre cost scaling is confirmed in the CNN–MNIST construction and in a frozen-backbone adapter-plus-classifier ResNet construction. Transverse response amplification is confirmed in a trainable layer4-plus-classifier system. A frozen GPT-2 rank-2 LoRA domain adds cross-domain evidence: selected tangent value is confirmed on codeword prompts, while current-versus-fixed cost separation is prospectively

confirmed on disjoint natural-English development and response continuations. The restricted F16 ordering and the action-cost separation remain confined to TinyCNN/MNIST. The sixteen seeds in each ResNet cohort vary training, not the dataset, pretrained representation or architecture.

11.2 Geometric scope

The transported chart is eight-dimensional throughout: a controlled subchart of the response kernel, not the complete kernel, whose numerical dimension in the CNN construction is 4146. The complete ResNet backbone is never adapted, the GPT-2 audits are confined to a rank-2 LoRA-B block of the last two layers rather than the complete parameter space, no global smooth fibre bundle is established, and nothing is claimed across rank-changing or singular regions. Natural language-model training trajectories are not tested. On a constant-rank region $\ker DR$ is already the level-set tangent distribution, so its local integrability is not what is at issue. The capacity H_{cap} is declared rather than derived, and its sensitivity to R , Q , the ridge, the capacity floor and the probe set has not been mapped.

One geometric limit concerns the unit in which every effect in Section 3 is measured. Since $\dim \ker DR(\theta) = N - \text{rank } DR(\theta)$, the fraction of any realized update that is response-preserving is set by the declared anchor set, a free design parameter. A small anchor set leaves most of every update inside the fibre; a large one leaves little there. No sweep over anchor count, kernel dimension or realized tangent share accompanies those results, so the defensible form of the finding is the ordering, not the magnitude.

11.3 Low-response Pareto scope

The Pareto audits use finite α grids and four frozen response budgets rather than continuous constrained optimization. The reduced ResNet construction is not the full training configuration. The GPT-2 audit is confined to rank-2 native-LoRA-B coordinates in layers 10 and 11 and an artificial color-code task expressed through disjoint natural-language templates. The result does not establish full-parameter language-model geometry, general semantic invariance or optimizer naturality.

11.4 Variational scope

The action ordering is an argmin over the frozen six-algorithm family $\mathcal{A}_6$ at four finite radii. It is not a statement about arbitrary causal paths, an Euler–Lagrange condition, or a global minimality theorem. The arms share a functional but not a parameter endpoint; the Newton-retraction Fisher length is an operational proxy, not the unique notion of realizability cost; four radii do not establish an exact continuum limit; the leading coefficient or renormalized density of the vanishing realizability cost remains uncertified; and no optimizer is shown to select the current-kernel transported chart.

11.5 Statistical and protocol scope

The confirmations have different margins and should not be summarized by the tightest of them. The cost-scaling cohort passed 15/16; the F16 cohort passed 14/16, exactly meeting its gate and leaving no eligibility margin; the transverse cohort passed 16/16 against a frozen 12/16. Two Fisher cohorts remain formally ineligible and are recorded as such in Supplementary Section SA. The Fisher metric may change with chart basis, sampling distribution, damping and response definition. The dictionary underlying Section 3 is finite and demonstrably incomplete, several diagnostic arms use oracle tangent norms, the low-dimensional geometric audits use three-parameter models in which many endpoint-matched paths are close by construction. The six-regime comparison is descriptive rather than a population-level statistical claim; the complete finite-family ordering and its frozen gates are reported in the Supplementary Information. Separate GPT-2 soft-prompt distance audits in Supplementary Section SA test source-frozen Fisher length only and make no result here cross-architecture.

12 Conclusion

The evidence supports one state-dependent mechanism with two experimentally distinct consequences. First, response-preserving freedom moves with model state: source-fixed replay incurs different finite realizability cost from current-kernel transport, and small chart drift can be transversely amplified in active response space. Second, the instantaneous kernel can provide locally usable stability–plasticity freedom. Under prospectively frozen response budgets, it yields greater held-out learning utility than recorded-optimizer, source-kernel, time-shifted and signed-permuted controls in 7/8 reduced ResNet seeds and 8/8 GPT-2 native-LoRA-B seeds, with all 24 noninitial GPT-2 nodes passing the frozen ordering.

The geometric evidence behind the first consequence is unchanged. Selected response-preserving directions are not inert redundancy. Their adaptive-value ordering survives prospective confirmation in the complete final-classifier space of a frozen ResNet-18 representation, with all sixteen seeds passing across three anchor counts. More importantly, the CNN and ResNet audits establish the central moving-fibre mechanism: transporting a direction in the current response kernel and replaying a direction frozen at the source produce systematically different finite realizability-cost scaling. In CNN–MNIST, all fifteen eligible seeds passed, with $\alpha_{\text{moving}} = 0.684$ against $\alpha_{\text{fixed}} = 0.007$ and a median cost ratio of 21.30; the independent ResNet-18/CIFAR-10 confirmation passed 16/16 seeds. A trainable ResNet block further shows why this distinction matters: current-fibre and source-fixed directions remain nearly collinear, yet their active response residuals differ by more than three orders of magnitude. The relevant object is therefore not merely a large null space, but a moving null distribution embedded in a strongly anisotropic response geometry.

The restricted action audit also fixes the present theoretical boundary. The constructed true-natural-gradient path minimizes the declared Moving-Fibre F16 action among six causal online algorithms at every tested radius, but it is systematically not the least expensive path to realize, paying a median 2.25-fold correction-cost premium. Thus moving-fibre geometry, path selection and path realization cannot yet be identified with a single global variational functional. The result constrains any future action principle rather than proving one.

The earlier GPTW-v3 result remains an independent cross-modal confirmation of current-versus-fixed correction-cost scaling, with 8/8 new seeds and 32/32 interior nodes of that protocol, finest-radius fixed-to-current correction-cost ratios of 8.87–35.56, and current and fixed exponents $\alpha_{\text{current}} = 1.9987\text{--}2.0040$ and $\alpha_{\text{fixed}} = 1.0625\text{--}1.2106$ reported as numerical-correctness checks. It should not be conflated with the 24-node low-response Pareto audit.

The strongest supported conclusion is therefore that response-preserving freedom is a moving geometric object whose stale use has finite and amplifiable consequences, and whose instantaneous kernel defines a superior low-response learning frontier within the declared ResNet and GPT-2 adapter charts. Neither result implies that ordinary optimizers navigate this frontier naturally; ordinary optimizer navigation is not established. The restricted action-cost failure also precludes identifying the mechanism with a single global variational law.

In this restricted sense, response-fibre geometry links behaviour and learning: models that agree on a declared response can retain different locally accessible capacities for subsequent learning, and those capacities change as the response fibre moves. Behaviour here remains response-relative and finite, and no claim is made about complete functional equivalence or natural optimizer navigation.

Data and code availability

Three result packages must be registered separately.

(1) GPTW-v3 correction-cost package (publicly archived). Protocols, executable programs, frozen gates, per-seed records and checksums are archived at <https://github.com/papasop/neural-fibre-geometry> commit 236f646c472018a7e38be11fd658519763bc2346.

No release tag or DOI has been assigned to this version. The GPT-2 r3 and v2 evidence is present under

evidence/gptw_response_fibre_v1_1_0/, including frozen protocols, reports, per-seed JSON files, run.py, SHA256SUMS, claim boundaries, development records and snapshot verification.

(2) Reduced ResNet v4.6 low-response Pareto package. (3) GPT-2/native-LoRA-B low-response Pareto eight-seed package. The ResNet v4.6 and GPT-2/native-LoRA-B low-response Pareto packages are not contained in commit 236f646c472018a7e38be11fd658519763bc2346 and are therefore not yet independently reconstructible from the cited public snapshot. They are held locally and are not described here as publicly archived.

At this commit, the ResNet v4.1b, v4.2c, v4.2c-r1 and v4.2d evidence packages cited in Sections 5.3–6 are likewise not in the public archive; those numerical claims therefore cannot yet be independently reconstructed from this snapshot. Availability statements refer only to records actually contained in the cited commit.

References

1. S. Ainsworth, J. Hayase, and S. Srinivasa. Git Re-Basin: merging models modulo permutation symmetries. *International Conference on Learning Representations*, 2023.
2. S. Amari. Natural gradient works efficiently in learning. *Neural Computation*, 1998.
3. Y. Bengio, A. Courville, and P. Vincent. Representation learning: a review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8):1798–1828, 2013.
4. L. Dinh, R. Pascanu, S. Bengio, and Y. Bengio. Sharp minima can generalize for deep nets. *Proceedings of the 34th International Conference on Machine Learning*, PMLR 70:1019–1028, 2017.
5. F. Draxler, K. Veschgini, M. Salmhofer, and F. A. Hamprecht. Essentially no barriers in neural network energy landscape. *International Conference on Machine Learning*, PMLR, 2018.
6. M. Farajtabar, N. Azizan, A. Mott, and A. Li. Orthogonal gradient descent for continual learning. *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, PMLR 108:3762–3773, 2020.
7. J. Frankle, G. K. Dziugaite, D. M. Roy, and M. Carbin. Linear mode connectivity and the lottery ticket hypothesis. *International Conference on Machine Learning*, PMLR, pages 3259–3269, 2020.
8. T. Garipov, P. Izmailov, D. Podoprikin, D. P. Vetrov, and A. G. Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. *Advances in Neural Information Processing Systems 31*, 2018.
9. R. Hecht-Nielsen. On the algebraic structure of feedforward network weight spaces. In *Advanced Neural Computers*, pages 129–135. Elsevier, 1990.
10. S. Hochreiter and J. Schmidhuber. Flat minima. *Neural Computation*, 9(1):1–42, 1997.
11. E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.
12. G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and A. Farhadi. Editing models with task arithmetic. *International Conference on Learning Representations*, 2023.
13. A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. *Advances in Neural Information Processing Systems 31*, 2018.
14. N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang. On large-batch training for deep learning: generalization gap and sharp minima. *International Conference on Learning Representations*, 2017.
15. G. Saha, I. Garg, and K. Roy. Gradient projection memory for continual learning. *International Conference on Learning Representations*, 2021.

16. D. Soudry, E. Hoffer, M. S. Nacson, S. Gunasekar, and N. Srebro. The implicit bias of gradient descent on separable data. *Journal of Machine Learning Research*, 19(70):1–57, 2018.
17. G. Zeng, Y. Chen, B. Cui, and S. Yu. Continual learning of context-dependent processing in neural networks. *Nature Machine Intelligence*, 1(8):364–372, 2019.
18. C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals. Understanding deep learning requires rethinking generalization. *International Conference on Learning Representations*, 2017.